

Scottish Church College

Physics GNUPLOT Practical problem set

1. (i) Define a function

$$y = f(x) = x^5 + 2x^2 + 6.$$

Set range of x values from -10 to 10 . Within this range plot above function. Both the axes must be shown on the plot. The label of x axis and y axis will be “ x values” and “new function” respectively. Ticks of the x axis will be at interval 2 . Store the output in a file and print it.

- (ii) Define a function

$$y = f(x) = x^2 \sin(x).$$

Set range of x values from -12 to 12 . Within this range plot the above function. Both the axes must be shown on the plot. The label of x axis and y axis will be “ x values” and “my function” respectively. Ticks of the x axis will be at interval 2 . Store the output in a file and print it.

- (iii) Plot $\sin(x)$ and $2\sin(-x)$ simultaneously on the same plot, within the range -3π to 3π . Use the system defined value of 3π as $3 * \text{pi}$. The label of x axis and y axis will be “ x values” and “standing wave” respectively. Store the output in a file and print it. Move the mouse pointer on the nodes created on the plot and note down the values on paper.

- (iv) Define the following function:

$$f(x) = \begin{cases} x, & -2 \leq x \leq 2, \\ 0.5, & \text{otherwise.} \end{cases}$$

Set range of x values from -10 to 10 and range of y values (y is treated as $f(x)$) from -5 to $+5$. Keep ticks in x axis at interval 2 and ticks in y axis at interval 1 . Set x label as “ x value” and y label as “jerk function”. Plot the function, store it in a file and print it.

- (v) Define three functions in the given range:

$$\begin{aligned} f_1(x) &= x, & x > 4, \\ f_2(x) &= -x^2, & -4 < x < 4, \\ f_3(x) &= -x, & x < -4. \end{aligned}$$

Set the range of x values from -10 to $+10$ and the range of y values (y is treated as $f(x)$) from -4 to $+10$. Set ticks of the x axis at the interval 1 . Set the label of x axis as “ x values” and the label of y axis as “piecewise continuous functions”. Plot the three functions, store the output in a file and give a printout.

- (vi) Set a function

$$f(x) = 2x^3 + 9x - 11.$$

Set range of x values from -15 to $+10$ and range of y values (y is treated as $f(x)$) from -7000 to $+2200$. Plot this function in the range. Indicate y axis on the graph. The x label will be “ x value” and y label will be “new function”. Set the tics in the x axis at interval 2.5 . Store the output in a file and print it.

- (vii) Set a function

$$f(x) = \cos(x) - 0.5x.$$

Set range of x values from -20 to $+20$ and range of y values (y is treated as $f(x)$) from -10 to $+12$. Plot this function in the range. Show the x axis on the graph. The x label will be “ x value” and y label will be “ $\cos(x) - 0.5x$ ”. The tics in the x axis will be at interval of 2 . Store the output in a file and print it.

(viii) Define the following function:

$$f(x) = \begin{cases} x, & x \geq 2, \\ x^2, & -3 \leq x \leq 2, \\ -x, & x \leq -3. \end{cases}$$

Set range of x values from -8 to 8 and range of y (y is treated as $f(x)$) values from -2 to 12 . Keep ticks in x axis at interval 2 . Set x label as “ x value” and y label as “jerk function”. Plot the function, store it in a file and give a printout.

(ix) Define three functions in the given range:

$$\begin{aligned} f_1(x) &= x, & x > 2, \\ f_2(x) &= x^2, & -1 < x < 2, \\ f_3(x) &= 2, & x < -1. \end{aligned}$$

Set the range of x values from -6 to $+6$ and the range of y values (y is treated as $f(x)$) from -2 to $+7$. Set tick label of the x axis at interval of 1 . Set the label of x axis as “ x values” and the label of the y axis as “piecewise continuous functions”. Plot the three functions, store the output in a file and give a printout.

(x) Plot $\sinh^2(x)$, $-\cosh(x)$ and $-\{1 + \tanh(x)\}$ on same graph. The line width of the curves will be gradually increasing. That is, the line width of $-\cosh(x)$ will be thicker than $\sinh^2(x)$ and the line width of $-\{1 + \tanh(x)\}$ will be thicker than $-\cosh(x)$. Set the range of the x values from -6 to $+6$ and the range of the y values from -8 to $+8$. The x axis label will be “ x values” and that of the y axis will be “hyperbolic functions”. Store the output in a file and give a printout.

(xi) Define a function

$$y = \ln(x^2).$$

Set the range of x values from -2 to $+2$ and the range of y values from -3 to $+3$. Show both the axis on the plot. The labels of x axis and y axis will be “ x values” and “ y values” respectively. Ticks of the x axis will be at interval of 0.5 . Ticks of the y axis will be at interval of 1 . Plot the function and store the plot in a file and print it.

(xii) Plot

$$\frac{\sin x}{x}, \quad \frac{\sinh x}{x},$$

and $y = 1$ on same graph. The line width of the three curves to be different. Set the range of x values from $-\pi$ to $+\pi$ and the range of y values from 0 to 5 . Show y axis on the plot. Set title “Plot of different functions of x ”. The labels of x and y axis to be “ x values” and “ y values” respectively. Store the plot in a file and print it.

(xiii) Define the following function:

$$f(x) = \begin{cases} x \cos x, & x < 0, \\ x^2 + 3x, & 0 \leq x \leq 1, \\ \ln x + 4, & x > 1. \end{cases}$$

Set the range of x values from -3 to 3 and the range of y values from -2 to $+5$. Show both the axes on the plot. Set title “Continuous function without derivative at all points”. Plot the function, store it in a file and print it.

(xiv) Define three functions in the given range:

$$\begin{aligned} f_1(x) &= \ln\left(\frac{1}{x}\right), & x > 0.1, \\ f_2(x) &= 3x^2 \sin x, & -1 < x < 0.1, \\ f_3(x) &= \cos 3x, & x < -1. \end{aligned}$$

Set the range of x values from -3 to $+3$ and the range of y values from -2.5 to $+2.5$. Show both the axes on the plot. Set y label “Piecewise continuous function”. Set ticks of the y axis at an interval of 0.5 . Plot the three functions and store the plot in a file and print it.

- (xv) Define a function

$$y = x \sin 3x.$$

Set the range of x values from -3 to $+3$ and the range of y values from -4 to $+4$. Show both the axes on the plot. The labels of x axis and y axis will be “ x values” and “ y values” respectively. Ticks of the x axis will be at interval of 0.5 . Ticks of the y axis will be at interval of 1 . Plot the function and store the plot in a file and print it.

- (xvi) Plot

$$2x \sin 2x, \quad \frac{1}{2}x \sinh\left(\frac{x}{2}\right)$$

on the same graph. The line width of the two curves to be different. Set the range of x values from $-\pi$ to $+\pi$ and the range of y values from -5 to 5 . Show both the axes on the plot. Ticks of x axis will be at interval of 0.5 . The label of x and y axis to be “ x values” and “ y values” respectively. Store the plot in a file and print it.

- (xvii) Define the following function:

$$f(x) = \begin{cases} x^2, & x < 1, \\ x, & 1 \leq x \leq 2, \\ \frac{x^3}{4}, & x > 2. \end{cases}$$

Set the range of x values from 0 to 3 and the range of y values from 0 to 8 . Ticks of x axis will be at interval of 0.5 and that of y axis will be at interval of 1 . Set title “Polynomials of x ”. Plot the function and store it in a file and print it.

- (xviii) Define three functions in the given range:

$$\begin{aligned} f_1(x) &= \frac{1}{x}e^x, & x > 1, \\ f_2(x) &= x^2 \tanh x, & -1 < x < 1, \\ f_3(x) &= -\frac{1}{x}e^{-x}, & x < -1. \end{aligned}$$

Set the range of x values from -3 to $+3$ and the range of y values from -7 to $+7$. Show both the axes on the plot. Set y label “Piecewise continuous function”. Set ticks of the y axis at an interval of 1 . Plot the three functions and store the plot in a file and print it.

- (xix) Three functions are defined in the given range as the following :

$$\begin{aligned} f(x) &= 1.45x^3 - 6\sqrt{x}, & x \geq 2 \\ g(x) &= \sin^2(x), & -1 \leq x < 2 \\ h(x) &= |x|e^x, & x < -1 \end{aligned}$$

Set the range of x values from -2 to $+3$ and the range of y values from -5 to $+5$. Set ticks of the x -axis at an interval of 1.0 and ticks of y -axis at an interval 2.0 . Set title Piecewise continuous functions’. Plot the three functions using GNUPLOT and store the plot in suitable format.

- (xx) Plot the following functions for $0 \leq x \leq \pi$ on the same graph using GNUPLOT. Set 1000 samples. Mark each curve with different line style and line widths. Label x -axis and y -axis as ‘ x -values’ and ‘ y -values’ respectively. Set the title of the plot ‘Function plots’. Save the plot in a suitable format.

$$f_1(x) = \sin^2(\sqrt{x}), \quad f_2(x) = e^{-x/2}, \quad f_3(x) = 12.34x^{5/3}$$

- (xxi) Plot the following functions for $-10 \leq x \leq 10$ on the same graph using GNUPLOT. Set the range of y-values from -4 to $+4$. Mark each curve with different line style and 1000 samples, line widths, Label x-axis and y-axis as 'x-values' and 'y-values' respectively. Set the title of the plot Save the plot in a suitable format. Function plots'.

$$f_1(x) = \coth(x) - \frac{1}{x}, \quad f_2(x) = e^{-\sin(x)}, \quad f_3(x) = x|x|$$

2. (i) Make the following data table:

x	y
-2.7	-15.07
-1.5	-9.5
1.4	3.72
3.3	12.5
6.1	25.2

Plot the data in the second column against that of the first column. Set the range of x values from -4.0 to $+8.0$. Try to fit the data with the equation of type $y = mx + c$. Show the points and the fitting line on the plot. Arrange the range of y values on the axis in such a way that the value of c can be realized from the axis. Save the fitted plot in a file and give a printout. Mention the values of m and c coming after fitting the data.

- (ii) Make the following data table:

x	y
-5.1	21.8
-3.4	16.12
-0.5	7.05
1.57	0.42
6.2	-14.2

Plot the data in the second column against that of the first column. Set the range of x values from -7.0 to $+7.0$. Try to fit the data with the equation of type $y = mx + c$. Show the points and the fitting line on the plot. Draw y axis on the plot. Arrange the range of y values on the axis in such a way that the value of c can be realised from the axis. Save the fitted plot in a file and give a printout. Mention the values of m and c coming after fitting the data.

- (iii) Make the following data table:

x	y
2.1	-2.5
4.6	7.7
5.7	12.17
6.6	15.9
8.2	22.4

Plot the data in the second column against that of the first column. Set the range of x values from 0 to $+9.0$. Try to fit the data with the equation of type $y = mx + c$. Show the points and the fitting line on the plot. Draw y axis on the plot. Arrange the range of y values on the axis in such a way that the values of c can be realised from the axis. Save the fitted plot in a file and give a printout. Mention the values of m and c coming after fitting the data.

- (iv) Make the following data table in a file:

x	y
-2.0	-4.15
-1.3	-2.40
1.8	6.60
2.5	8.98
3.6	11.94

Plot the data in second column against that of the first column. Set the range of x values from -3 to 4 . Try to fit the data with the equation of type $f(x) = mx + c$. Mention the values of m and c after fitting the data. Show the points and the fitted line on the same plot. Draw both the axes on the plot. Arrange the range of y values on the axis such that the value of c can be realized from the axis. Store the plot in a file and print it.

- (v) Make the following data table in a file:

x	y
0.5	0.21
0.83	1.05
1.16	2.40
1.82	9.01
2.81	30.75

Plot the data in second column against that of the first column. Set the range of x values from 0 to 3 and the range of y values from 0 to 35 . Try to fit the data with the equation of type $f(x) = ax^b$. Mention the values of a and b after fitting the data. Show the points and the fitted line on the same plot. Set title “Power series fit”. Store the plot in a file and print it.

- (vi) Make the following data table in a file:

x	y
-2.9	-5.96
-1.4	-3.81
0.8	-1.10
1.7	0.21
2.5	1.30

Plot the data in second column against that of the first column. Set the range of x values from -3 to 3 . Try to fit the data with the equation of type $f(x) = mx + c$. Mention the values of m and c after fitting the data. Show the points and the fitted line on the same plot. Draw both the axes on the plot. Arrange the range of y values on the axis such that the value of c can be realized from the axis. Store the plot in a file and print it.

- (vii) Make the following data table in a file:

x	y
-2.8	14.65
-1.2	5.81
0.7	4.48
1.8	8.33
2.9	15.56

Plot the data in second column against that of the first column. Set the range of x values from -3 to 3 and the range of y values from 0 to 16 . Try to fit the data with the equation of type

$$f(x) = ax^2 + b.$$

Mention the values of a and b after fitting the data. Show the points and the fitted curve on the same plot. Set title “Polynomial fit”. Store the plot in a file and print it.

- (viii) Save the following data table in a file. Fit the data points of the second row against that of

x	-3.0	-2.0	-1.0	0.0	1.0	2.0	3.0	4.0
$f(x)$	3.422	4.057	4.808	5.7	6.756	8.008	9.492	11.251

the first row using GNUPLOT with the equation $f(x) = ae^{bx}$. Write the values of the fitted parameters a, b . Show the fitted curve along with the data points in the same plot. Set the

title "Data fitting" and 'x' and 'f(x)' for x-label and y-label respectively. Save the plot in a suitable format.

- (ix) Create the following data table in a file:

x	0.0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
$f(x)$	0.0	0.006	0.053	0.195	0.49	1.0	1.792	2.935	4.5	6.56	9.19

Fit the above data table with $f(x) = x^a$ with parameter a . Write the value of a . Plot the above data table along with $f(x)$.

Set the title 'Data fitting' and 'x' and 'f(x)' for x-label and y-label respectively. Save the plot in a suitable format.

- (x) Save the following data table in a file. Fit the data points of the second row against that of the

x	1.22	1.44	1.67	1.89	2.11	2.33	2.56	2.78	3.0
$f(x)$	5.013	3.65	2.78	2.19	1.77	1.467	1.234	1.054	0.910

first row using GNUPLOT with the equation $f(x) = a^x$. Write the values of the fitted parameters a, b . Show the fitted curve along with the data points in the same plot. Set the title "Data fitting" and 'x' and 'f(x)' for x-axis and y-axis respectively.

3. (i) Show the polar plot of the functions $\sin^3(t)$ and $\cos^2(2t)$ where t stands for a parameter, the angle. Put horizontal axis label as "t value" and vertical axis label as "the range". Put a label "polar plot" at position $(0, -0.7)$. Set the vertical axis passing through $t = 0$. Save the plot in a file and give a printout.
- (ii) Show the polar plots of the functions $\tan(6t)$ and $\cot(6t)$ where t stands for angle. Put horizontal axis label as "t value" and vertical axis label as "the range". Set the ranges of both the axes from -20 to $+20$. Put a label " $\tan(6t), \cot(6t)$ " at the position $(4, 12)$. Set the horizontal and vertical axes passing through $t = 0$ and $\text{range} = 0$. Save the plot in a file and give a printout.
- (iii) Show the parametric plot of the functions

$$x = 2 \cos^3 t, \quad y = 2 \sin^3 t,$$

where t is the parameter. Set the range of t values from $-\pi$ to π . The labels of x and y axis will be " $x = 2 \cos^3 t$ " and " $y = 2 \sin^3 t$ ". Set title "Astroid". Draw both the axes on the plot. Store the plot in a file and print it.

- (iv) Show the polar plot of the function

$$r = t,$$

where t is the angle made with the reference direction (positive x axis). Set samples to 10000, set range from 0 to 6π . Show both the axes on the plot. The labels of x and y axis will be " $x = r \cos t$ " and " $y = r \sin t$ ". Set title "Spiral". Store the plot in a file and print it.

- (v) Show the parametric plot of the functions

$$x = 2 \sin t, \quad y = 2 \sin 2t,$$

where t is the parameter. Set the range of t values from $-\pi$ to π . The label of x and y axis will be " $x = 2 \sin t$ " and " $y = 2 \sin 2t$ ". Set title "Lissajous Figure". Draw both the axes on the plot. Store the plot in a file and print it.

- (vi) Show the polar plot of the functions

$$r = 4 + 3 \cos t, \quad r = 3 + 4 \cos t,$$

where t is the angle made with the reference direction (positive x axis). Show both the axes on the plot. The labels of x and y axis will be " $x = r \cos t$ " and " $y = r \sin t$ ". Set title "A pair of Limacon". Put label "polar plot" at left corner at $(-6, 6)$. Store the plot in a file and print it.